# Arduino Servo Motor Control

Building Smart Gates & Precise Angular Control | Student Lesson Guide

## 1. Introduction: How Does a Toll Gate Lift Up?

Have you ever noticed how automatic toll barriers at FASTag gates or residential security gates lift up to let cars pass? The barrier arm always lifts to an exact **90° angle**, stops firmly in mid-air, and then lowers back down to **0°** once the vehicle passes.

How does a machine achieve such precise positioning every single time? The answer lies in a special component called a **Servo Motor** controlled by a micro-controller like **Arduino UNO**.

## 2. What is a Servo Motor?

A **servo motor** is a rotary actuator that allows for precise control of angular position, velocity, and acceleration. Unlike standard motors, it knows its exact position and holds it firmly.

| Servo Motor | Standard DC Motor |
|---|---|
| Moves to a specific target angle (e.g., 0° to 180°). | Spins continuously in a circle as long as power is applied. |
| Holds its position firmly against external resistance. | Cannot hold a precise position or fixed angle. |
| Ideal for robotics, smart gates, steering mechanisms. | Ideal for wheels, fans, and simple spinning mechanisms. |

## 3. Meet the SG90 Servo Motor & Wiring Guide

The **SG90** is a lightweight, high-output micro servo motor popular in educational electronics. It operates on 3 wires:

| Wire Color | Function | Arduino UNO Connection | Mnemonic / Memory Trick |
|---|---|---|---|
| 🔴 **Red** | VCC (Power Supply) | **5V Pin** | Red = Power / Energy |
| 🟤 **Brown / Black** | GND (Ground / Earth) | **GND Pin** | Brown/Black = Earth / Ground |
| 🟡 **Yellow / Orange** | Signal (PWM Control) | **Digital Pin 9** (PWM) | Yellow = Traffic Signal (Control) |

> ⚠️ **Spot the Wiring Mistake!**
> Always double-check your connections before powering on your Arduino! Common mistakes include swapping the Signal and VCC lines or plugging the power line into a 3.3V pin instead of 5V. Incorrect wiring can prevent the motor from moving or cause erratic behavior.

## 4. Writing Arduino Code to Control the Servo

To control a servo motor in Arduino, we use the built-in `Servo.h` library. Below is a fundamental sketch to attach and command the servo to a specific position:

```
// Step 1: Include the Servo Library
#include <Servo.h>

// Step 2: Create a Servo Object
Servo myServo;

void setup() {
  // Step 3: Attach the signal pin to Digital PWM Pin 9
  myServo.attach(9);
}

void loop() {
  // Step 4: Command servo to move to 90 degrees
  myServo.write(90);
  delay(1000); // Wait 1 second for the motor to reach position
}
```

### Simulating a Smooth Servo Sweep (0° to 180°)

To sweep the servo arm smoothly across its range, we loop from 0° up to 180° and back, using small delays between steps so the motor has time to reach each angle:

```
void loop() {
  // Sweep from 0 to 180 degrees
  for (int pos = 0; pos <= 180; pos += 1) {
    myServo.write(pos);
    delay(15); // Wait 15ms for smooth motion
  }

  // Sweep back from 180 to 0 degrees
  for (int pos = 180; pos >= 0; pos -= 1) {
    myServo.write(pos);
    delay(15);
  }
}
```

## 5. Real-World Applications in India & Beyond

Servo motors are integral components in modern automation and engineering:

- **FASTag Toll Barriers:** Lifts the barrier arm exactly 90° when a valid tag is scanned and closes it after the vehicle passes.
- **Residential Society Boom Barriers:** Automates security entry and exit gates for gated communities.
- **ISRO Space Technology:** Solar panels on satellites and rover robotic arms rely on precise servo actuators to deploy and position instruments accurately in space.

### 🚀 DIY Hands-On Challenge: Build Your Smart Gate Arm!

**Materials Needed:** SG90 Servo Motor, Arduino UNO, Jumper Wires, Popsicle Stick, Double-sided tape.

1. Fix a popsicle stick to the plastic servo horn/arm using double-sided tape or Blu-tack.

2. Connect the servo wires to 5V, GND, and Pin 9 on the Arduino.
3. Write a program that opens the barrier to 90° when a car approaches, holds it open for 3 seconds, and then closes it back to 0°.

# Practice Questions & Assessment

Test your knowledge on Arduino Servo Motor Control. Answer all questions below.

**Question 1 — Multiple Choice**

**How many wires does an SG90 servo motor have, and what are their specific functions?**

A) 2 wires: Power (+) and Ground (-)
B) 3 wires: Red (Power 5V), Brown/Black (Ground), Yellow/Orange (Signal)
C) 4 wires: VCC, GND, Step, and Direction
D) 3 wires: Red (Signal), Black (5V Power), Yellow (Ground)

**Question 2 — Conceptual Comparison**

**Explain the main technical difference between a standard DC motor and a Servo motor. In what practical scenario would you choose a servo motor over a DC motor?**

**Question 3 — Arduino Programming**

**Which Arduino header file must be included at the top of your sketch to control a servo motor, and what method/function connects the servo object to a specific digital pin?**

**Question 4 — Code Analysis**

**Look at the following line of code: `mySservo.write(90);`. What physical action does this command cause the servo motor to perform? How long will it stay in that position if no other commands are given?**

**Question 5 — Troubleshooting & Wiring**

**A student connects an SG90 servo motor to an Arduino UNO, but when they run `myServo.write(90);`, the motor makes a buzzing sound and does not move properly. List two potential wiring or power issues that could cause this problem.**

**Question 6 — Real-World Engineering Challenge**

**Design an algorithmic flow (or write pseudo-code) for an automated FASTag toll gate system. Describe how the gate should behave from the moment a vehicle approaches to when it drives through and leaves. Mention the servo angles used for opening and closing.**